# School of Future Tech

**Case Study Report on**

**FlyPrivate: A High-End Charter Jet Marketplace**

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## Introduction

This project focuses on the Aviation Industry, specifically building a high-end platform for booking private charters. The goal was to create a digital marketplace where users can view aircraft details and receive instant notifications for "empty leg" flights. Unlike standard travel sites, this platform is designed with a "Dark Mode Luxury" theme to cater to premium clients.

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## Problem Statement

Private aviation operators and charter services interact with high-value customers daily. Without a proper digital system, managing flight schedules, aircraft availability, and customized pricing quotes becomes slow, opaque, and inefficient.

The objective of this case study is to create a high-end charter jet marketplace that allows users to:

- Browse fleet specifications: Access detailed data on aircraft range, cabin height, and seating capacity.
- Calculate real-time costs: Generate instant booking estimates based on membership tiers and flight duration.
- Receive instant notifications: Get immediate alerts for "empty leg" flight opportunities that would otherwise remain unbooked.
- Validate booking requirements: Ensure passenger counts align with specific aircraft safety and capacity limits.

This system helps organize complex aviation data and booking workflows in a simple, structured, and visually premium way.

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# Case Study Design

The application architecture is divided into three functional modules to ensure a seamless flow:

- Fleet Discovery: A structured data table displaying aircraft models, range, and seating capacity.
- Detailed Interaction: A modal-based system that allows users to view high-resolution imagery and specific cabin specs (e.g., catering, height) without leaving the main page.
- Booking Engine: A grid-based form designed to handle complex inputs such as membership tiers and flight duration.

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# Methods and Technology Used

The project utilizes a modern frontend stack to achieve its luxury requirements:

- HTML5: Used for the semantic structure of aircraft specification tables and user input forms.
- CSS3 (Advanced Styling): Implementation of the Sapphire (#3C507D) and Quicksand (#E0C58F) palette. For a neat polished appearance, it makes use of Flexbox/Grid layouts and glazing (frosted glass effects).
- JavaScript:
  - Calculation Algorithm: Logic to compute total costs based on hourly rates and membership discounts.
  - Data Validation: Scripts to ensure passenger counts stay within the aircraft's safety limits.
  - UI Controls: Functions to manage modal popups for aircraft previews.

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# Implementation Details

The implementation is organized into three core files to maintain a modular codebase:

1. **aircraft-list.html**: Contains the fleet marketplace and logic for "empty-leg" notification popups.
2. **booking.html**: Houses the interactive reservation form with live price updating.
3. **style.css**: The centralized stylesheet that defines the premium visual identity and responsive grid system.

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# Conclusion

**Results:**

- The platform successfully delivers instant pricing for flights, including luxury tax breakdowns.
- The "Sapphire" and "Royal Blue" color scheme effectively establishes a high-contrast, elite brand identity.

- Validation logic prevents the booking of flights that exceed specific aircraft capacity limits.

**Conclusion:**

The FlyPrivate marketplace demonstrates that combining clean code (HTML/JS) with sophisticated design can solve transparency issues in the luxury travel sector. The project provides a solid foundation for a real-world charter booking system.